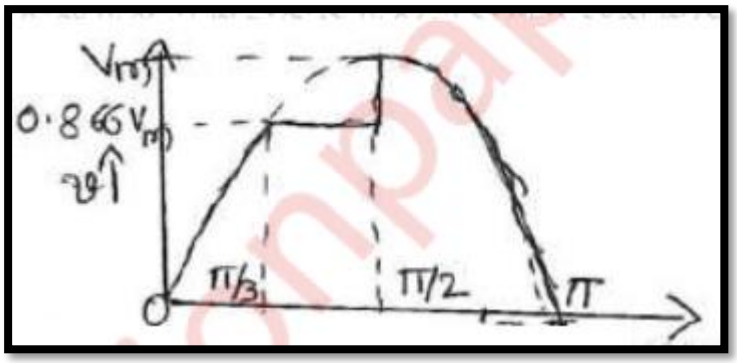
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BEEE Assignment Unit 2- AC circuits (2025-26)

1. Define RMS and Average value of an alternating current.
2. Define amplitude, cycle, period, instantaneous value, frequency, for single phase AC supply with suitable waveform
3. Explain the behaviour of pure Inductance in A C supply.
4. An alternating current of frequency 50Hz has a maximum value of 12A. Find the following: (i) Angular frequency in radians/sec;(ii) Equation of instantaneous value of current;(iii) Value of current after 5msec; (iv) RMS value of current and (v) Time taken to reach 9.6A for the first time.
5. Find the i) average value ii) rms value for the waveform given-



6. A circuit consists of a pure resistor and coil in series. Power dissipated in the resistor and in the coil are 1000 W and 250 W respectively. The voltage drops across the resistor and the coil are 200 V and 300 V respectively. Determine (i) value of pure resistance, (ii) resistance and reactance of the coil, (iii) combined resistance of the circuit, (iv) combined impedance, and (v) supply voltage

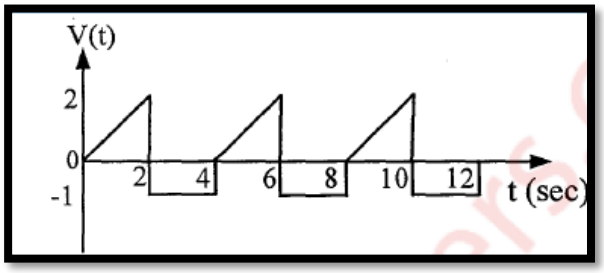
7. Two coils are connected in series across a 200 V, 50 Hz ac supply, The power input to the circuit is 2 kW and 1.15 kVAR. If the resistance and the reactance of the first coil are 5 W and 8 W respectively, calculate the resistance and reactance of the second coil. Calculate the active power and reactive power for both the coils individually

8. What is the necessary condition for resonance in series circuit? Derive the expression for resonance frequency

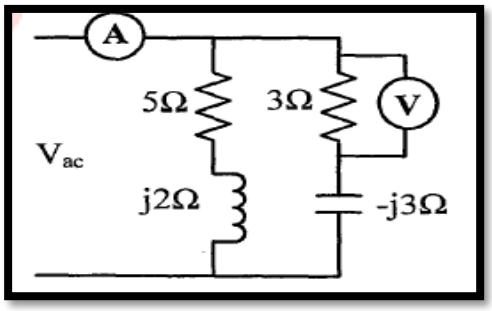
9. Coil A takes 2A at power factor of 0.8 lagging with an applied voltage of 10 V, second coil B takes 2 A with power factor of 0.7 lagging with applied voltage of 5V. What voltage will be required to produce a total current of 2 A with coil A and B in series. Find power factor in this case.

10. Derive the formula for resonant frequency of a series resonant circuit. State any five conditions of series resonance.

11. Find the Root Mean Square (RMS) value of the following waveform



12. In the parallel circuit, voltmeter across 3 Ω resistor reads 45 V. What is the indication on the ammeter? Also find the input power factor.



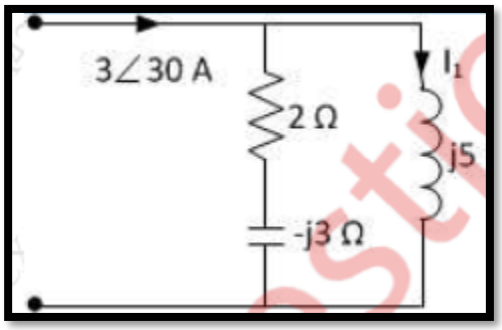
13. Two pure circuits elements in a series connection have the following current and applied voltage: V(t)=150sin(500t+10)v, i(t)=13.42sin(500t-53.3)A. Find the supply frequency(in Hz) and value of circuit elements.
14. Plot the variation of current, impedance, resistance, inductive reactance and capacitive reactance when supply frequency is varied in R-L-C series circuit**.**
15. Determine the relationship between phase and line voltage and current for star connected balanced load across a three-phase balanced system.
16. Determine the relationship between phase and line voltage and current for delta connected balanced load across a three-phase balanced system.
17. Three currents are meeting at a point. Find the resultant current.

$$i1 = 50\ \mathrm{Sin}(\omega t)A,\ i2 = 25\ \mathrm{Cos}(\omega t - \frac{\pi}{6})A,$$
$$i3 = -10\ \mathrm{Sin}(\omega t + \frac{\pi}{4})A.$$

18. If a voltage of v=200 (Sin 314t - 20°) V is being applied to an impedance. A current of 5<30°A flows through the circuit. Find resistance, reactance and PF of the circuit. (5<30 read this as 5angle 30 degree)

19. Find the current in inductor for the circuit shown

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20. A resistor of 20 Ω, inductor of 0.1 H and a capacitor of 50 μF are connected in series. A supply voltage 230 V, 50 Hz is connected across the series combination. Calculate the following: (i) impedance, (ii) current drawn by the circuit, (iii) phase difference and power factor, and (iv)active and reactive power and apparent power consumed by the circuit.
21. An alternating current of frequency 50 Hz has a maximum value of 12 A. Find (i) Equation of instantaneous value of current and (ii) Time taken to reach 9.6 for the first time.
22. A coil consumes 1 KW power from 50 Hz single phase ac supply and volt-ampere input is 2 KVA. If value of resistance of coil is 10 ohm then find the following parameters with respect to a coil (i) Value of inductance (ii) Phase angle (iii) Power factor (iv) Impedance (v) Voltage and Current equations (vi) KAR.
23. Three similar coils each having a resistance of 5 ohm and an inductance of 0.02 H are connected in star to a 440V, 50Hz, 3-phase supply. Calculate (i) line current and (ii) active and reactive power.
24. Two impedances of 12v + j16 ohm and 10 - j20 are connected in parallel across 230V, 50Hz, single phase ac supply. Find admittance, current and power factor of each branch,
25. State the necessary condition for resonance in a series circuit. Compare series and parallel resonance.

Course teachers
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